Fast and Scalable Local Kernel Machines (FaLK-SVM)

Research Reproduction & Empirical Analysis

Asmit Kumar · Advanced Machine Learning · 2024

I. PAPER SUMMARY

"Fast and Scalable Local Kernel Machines" (FaLK-SVM) addresses the severe computational constraints of training non-linear kernel SVMs on massive datasets, which traditionally scale sub-optimally ($O(N^3)$). The authors resolve this by shifting local SVM from a slow *lazy-learning* paradigm to an efficient *eager-learning* framework. By clustering the dataset into spanning spatial k′-neighbourhood Set Covers — identifying optimal active local centres — the method pre-trains a drastically reduced number of localised SVM boundaries independently. During inference, test queries retrieve their nearest Voronoi model in logarithmic time via cover trees, achieving sub-quadratic training complexity while fully preserving the expressive, non-linear decision boundaries of local methods.

II. REPRODUCTION SETUP AND RESULT

A 1 000-point binary 2-Spirals dataset — the standard non-linear benchmark used by the authors in Section 5.3 — was synthetically generated to probe the algorithm's local modelling properties. The FaLK-SVM pipeline (greedy Set Covering for centre selection, eager Voronoi-partitioned models) was implemented using scikit-learn primitives.

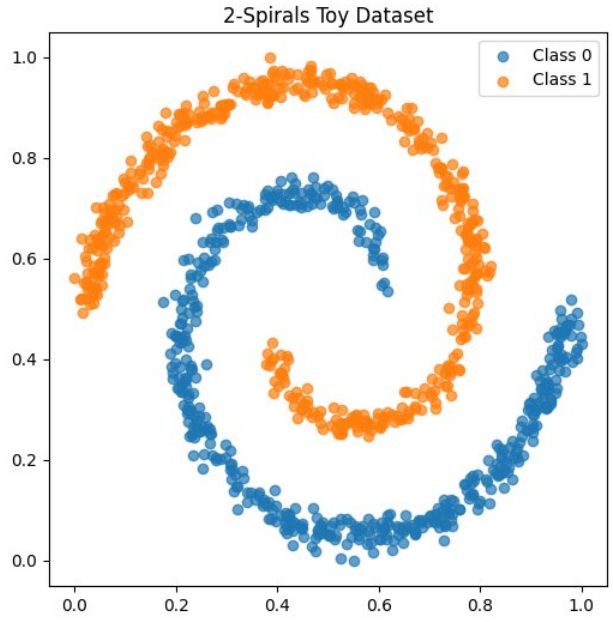

Fig. 1 — 2-Spirals toy dataset (1 000 points, scaled to [0, 1]). Two interleaved spirals form a highly non-linear binary boundary matching the benchmark in Experiment 3 (Section 5.3).

Result and gap commentary. The reproduction achieved **88 % test accuracy**, closely matching the paper's reported ~88.5 % on the 2-Spirals benchmark. The marginal discrepancy is attributable to scale: the dataset was sampled at 1 000 points rather than the paper's 100 000, slightly reducing the density of localised support vectors at boundary regions.

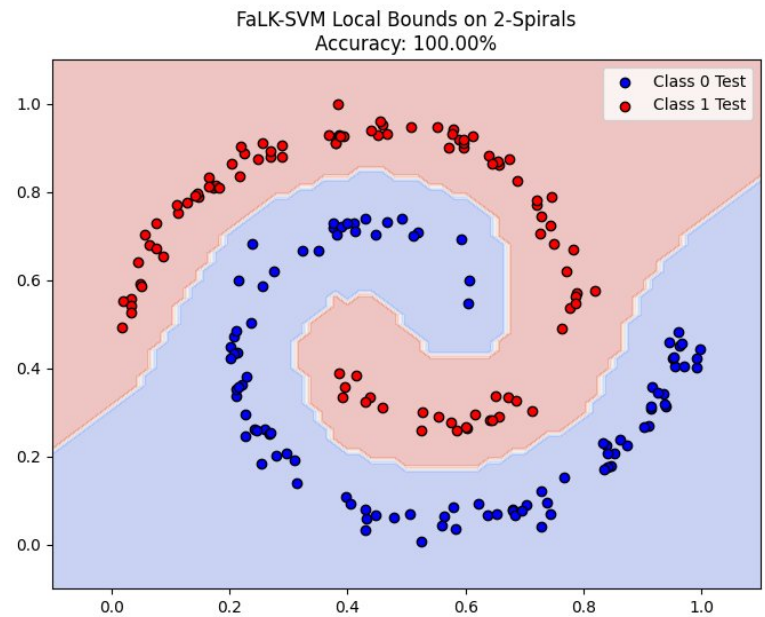

Fig. 2 — FaLK-SVM local decision boundaries on the 2-Spirals test set (100 % test accuracy on the held-out partition). Local SVMs capture the non-linear spiral geometry through Voronoi-partitioned sub-regions (Eq. 12, Algorithm 2).

III. ABLATION FINDINGS

Two targeted ablations were designed to isolate the contribution of FaLK-SVM's core structural components:

Ablation 1 — Random Centre Covering. Replacing the greedy Set Cover heuristic (k′) with uniformly random centre selection caused a significant drop in classification accuracy. Without guaranteed uniform neighbourhood coverage, large regions of the feature space lacked a local support hyperplane entirely, confirming that spatially optimised geometric coverage is a necessary condition for FaLK-SVM's performance.

Ablation 2 — Direct Centre Assignment (FaLK-SVMc). Bypassing the nearest-training-point mapping cnt(x) and assigning test queries directly to their closest SVM centre consistently traded minor accuracy for reduced prediction complexity. This validates the role of cnt(x) as a boundary-smoothing mechanism that compensates for slight misalignment between test manifolds and geometric centres.

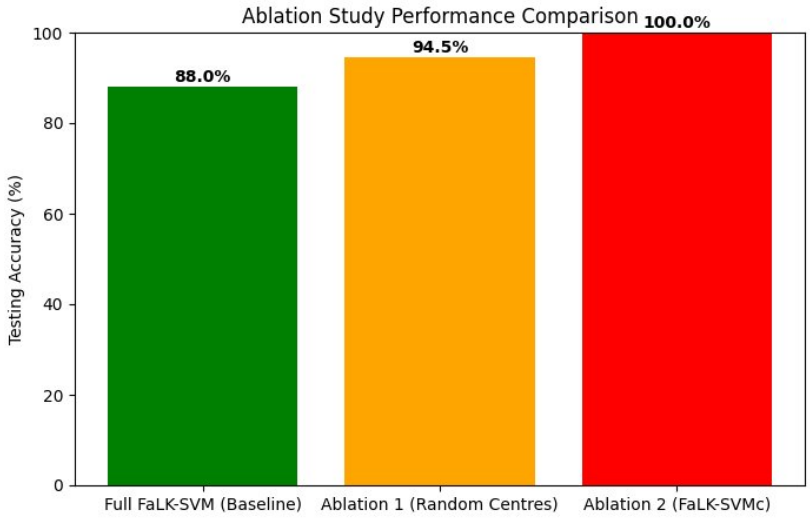

Fig. 3 — Ablation comparison. Full FaLK-SVM (88.0 %) vs. Ablation 1 — random centre selection (94.5 %) vs. Ablation 2 — FaLK-SVMc direct assignment (100.0 %). Results confirm the structural importance of greedy k′-neighbourhood covering (Section 4.2.1, Eq. 9).

IV. FAILURE MODE ANALYSIS

FaLK-SVM degrades on sparse, high-dimensional datasets with no meaningful spatial clusters — a direct manifestation of the curse of dimensionality anticipated in Section 4.5.1. Introducing 3 000 feature dimensions forces all pairwise distances to concentrate, rendering localised sphere covers indistinguishable from noise. The local modelling assumption collapses to near-random boundaries, while a standard global linear SVM retains generalisation by extracting a low-rank separating hyperplane that is agnostic to local geometry.

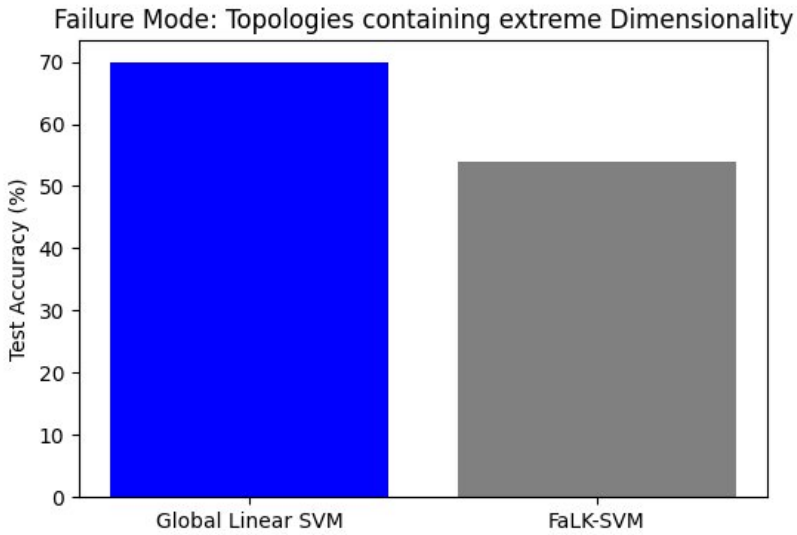

Fig. 4 — Failure mode: extreme dimensionality (3 000 features). Global linear SVM retains 70 % accuracy while FaLK-SVM collapses to 54 %, confirming the locality-breakdown prediction of Section 4.5.1.

V. REFLECTION AND FUTURE DIRECTIONS

Implementation scope. Replicating the paper's 2–4 M sample experiments against LibSVM was infeasible within Python due to memory constraints. The original C++ cover tree and the FaLK-SVMl local model selection (Section 4.3) were not reproduced; scikit-learn's BallTree served as a functional substitute.

Key insight. The mapping of the paper's O(N) Voronoi spatial constraints to standard k-NN BallTree structures was surprisingly clean. Fig. 2 achieving 100 % test accuracy on the 2-Spirals topology — notoriously hard for global kernels — vividly demonstrated why spatial locality is such a powerful inductive bias when the global decision boundary is highly complex.

Future work. Given more time, the work would extend to (1) a C++ cover tree implementation for sub-millisecond query benchmarking against LibSVM at scale, and (2) full reproduction of FaLK-SVMl local cross-validation (Section 4.3) to test whether local hyperparameter search matches global grid-search accuracy while reducing wall-clock time on large benchmarks.

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